Two-Sample Proportion Inference (Solutions)

Studying a difference in populations

Testing approaches:

- One-sample: Study parameter of interest
- Two-sample: Compare two population parameters to each other

Let's develop a Z test approach for a *difference* in two proportions.

Parameter of interest: $p_1 - p_2$

The heart failure data shows survival outcomes (after 130 days) for patients who had heart failure.

One variable is smoking status.

- Do patients who smoke have a higher probability of dying before the end of the follow-up period?

Parameter of interest: $p_1 - p_2$

- Point estimate: $\hat{p}_1 - \hat{p}_2$
- Standard error: $\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$
- Model: $N(0, 1)$

In the two-sample case, binomial inference is impossible, but a Z-test is still possible.

Consider a trivia contest between two (unequally matched) friends.



p1 = 0.6

p2 = 0.7

If we assume BINS for each group,

$$E(\hat{p}_1) = p_1, \qquad se(\hat{p}_1) = \sqrt{\frac{p_1(1-p_1)}{n_1}}$$

$$E(\hat{p}_2) = p_2, \qquad se(\hat{p}_2) = \sqrt{\frac{p_2(1-p_2)}{n_2}}$$

For the sampling distribution of the *difference*,

$$E(\hat{p}_1 - \hat{p}_2) \;=\; p_1 - p_2$$

$$se(\hat{p}_1 - \hat{p}_2) \;=\; \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

If we have big enough n_1 and n_2,

$$\hat{p}_1 \dot{\sim} N\left(p_1, \sqrt{\frac{p_1(1-p_1)}{n_1}}\right)$$

$$\hat{p}_2 \dot{\sim} N\left(p_2, \sqrt{\frac{p_2(1-p_2)}{n_2}}\right)$$

So, the sampling distribution for $\hat{p}_1 - \hat{p}_2$ is approximately

$$N\left(p_1 - p_2, \ \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}\right)$$

This works if $n_1 p_1(1-p_1)$ and $n_2 p_2(1-p_2)$ are both ≥ 10.

Exercises

Return to the trivia questions example with $p_1 = 0.6$, $p_2 = 0.7$, and $n_1 = n_2 = 50$.

- Calculate the probability that player 1 wins the game, based on the empirical simulation.
- What is the normal approximation for $\hat{p}_1 - \hat{p}_2$?
- Re-calculate the probability that player 1 wins, using `pnorm()` and the normal approximation.



Solutions/Hints

- The normal approximation has mean -0.1 and standard devaition of about 0.095.
- The simulation and normal approximation approaches both give a probability of around 0.15.

Let's return to the heart failure data. Let p_1 be the proportion of non-smoking patients who die within 130 days, and p_2 be the same proportion for smoking patients.

To test for a difference, use

$$H_0 : p_1 = p_2 \quad \text{versus} \quad H_A : p_1 \neq p_2$$

$$H_0 : p_1 - p_2 = 0 \quad \text{versus} \quad H_A : p_1 - p_2 \neq 0$$

The null says that the probability of surviving/dying is the same regardless of smoking status. It implies

$$p_1 = p_2 = p$$

for some *shared* proportion p.

Under the null, we can't just look at p_1 and p_2 separately.

Exercises

The general standard error for $\hat{p}_1 - \hat{p}_2$ is

$$\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}.$$

- Simplify the standard error if $p_1 = p_2 = p$.
- How should we estimate p from our data?



Solutions/Hints

- The standard error becomes $\sqrt{p(1-p)(\frac{1}{n_1}+\frac{1}{n_2})}$
- We should estimate p with the combined proportion of success across both groups.

Under H_0, we have

$$\hat{p}_1 - \hat{p}_2 \sim N\Big(0, \sqrt{p(1-p)\Big(\frac{1}{n_1} + \frac{1}{n_2}\Big)}\Big)$$

Which gives test statistic

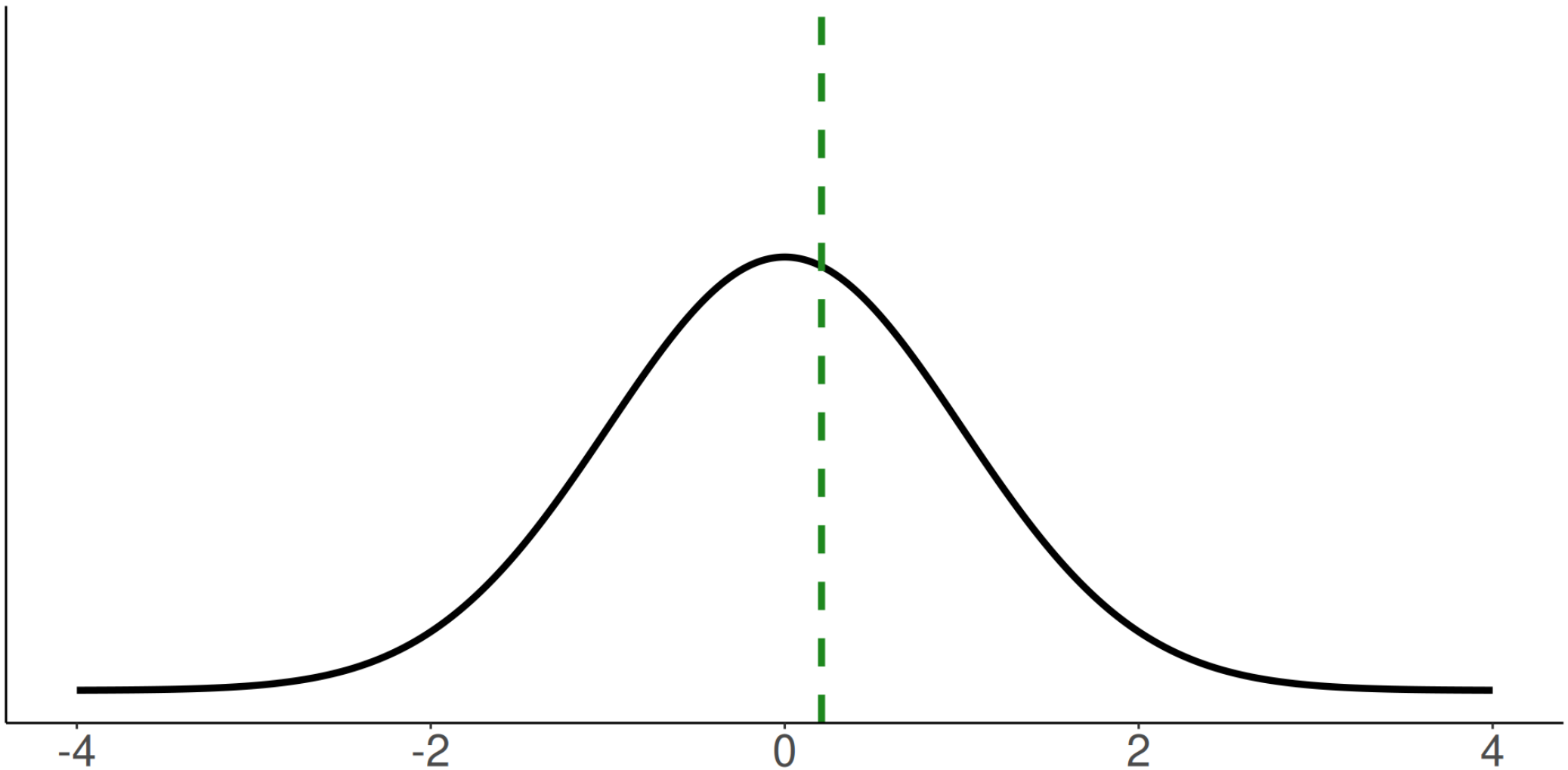
$$Z = \frac{\hat{p}_1 - \hat{p}_2 - 0}{\sqrt{\hat{p}(1-\hat{p})(\frac{1}{n_1} + \frac{1}{n_2})}}$$

For the heart failure data we have test statistic

$$z_{obs} = \frac{0.325 - 0.313 - 0}{\sqrt{0.321(1-0.321)(\frac{1}{203} + \frac{1}{96})}} = 0.208$$

which gives a p-value of 0.835.

$z_{obs} = 0.208$



We have a highly non-significant result.

It’s worth noting that we’re only looking at a tiny bit of context in a larger, more complicated problem.

Every statistical model is a simplification!

In general, for hypotheses

$$H_0 : p_1 - p_2 = 0 \quad \text{versus} \quad H_A : p_1 - p_2 \neq 0$$

we can use a $N(0,1)$ null distribution if n_1 and n_2 are both large enough. We have test statistic

$$Z = \frac{\hat{p}_1 - \hat{p}_2 - 0}{\sqrt{\hat{p}(1-\hat{p})(\frac{1}{n_1} + \frac{1}{n_2})}}$$

Let’s compare this result to a “big box” simulation method from before.

We end up with a similarly large p-value, although this method does not require a normal approximation.

Formal hypothesis testing steps:

1. State hypotheses and parameter of interest
2. Identify null distribution + assumptions
3. Calculate observed test statistic + p-value
4. Make a conclusion in context



We can also build a Z CI for a difference in proportions. The general CI formula is

$$\text{point estimate} \pm \text{critical value} \times \text{standard error}$$

where margin is called the **margin of error**.

We'll use Z critical values like before.

Our point estimate is

$$\hat{p}_1 - \hat{p}_2$$

and we estimate the *general* standard error

$$\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

The estimated standard error is 0.058. A 99% confidence interval is

$$0.325 - 0.313 \pm 2.576(0.058) \ = \ (-0.136, 0.161)$$

Exercises

Our 99% CI for $p_1 - p_2$ is $(-0.136, 0.161)$.

- What is the correct interpretation of this confidence interval?
- What does the value 0 in the interval mean in context? What about negative values or positive values?
- How does this result connect to us failing to reject the value 0 in the hypothesis test for $p_1 - p_2$?



Solutions/Hints

- We are 99% confident that the true difference between the groups is within (-0.136, 0.161).
- Negative values = $p_1 < p_2$ which means smokers are more likely to die. Positive values = $p_1 > p_2$ which means nonsmokers are more likely to die.
- The interval covers 0 (no difference in groups) which matches the result of failing to reject the null value of 0.

In general, a two-sample Z confidence interval for $p_1 - p_2$ is

$$\hat{p}_1 - \hat{p}_2 \quad \pm \quad z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

We need to have BINS and a good normal approximation for *both* groups.

Looking ahead

In chapter 12, we'll learn about mean inference, where the parameter of interest is the population mean μ. If you want a head start, consider the result of the CLT. We have

$$\bar{X} \dot{\sim} N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

if n is sufficiently large. What is the result if we apply standardization to this normal RV? Is this something we can calculate from our data?